

# Rahul Medicharla

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## TECHNICAL SKILLS

**Certifications:** AWS Solutions Architect Associate, Certified Kubernetes Application Developer (CKAD)

**Languages:** Go, Python, Javascript, Java, SQL, C#, C

**Frameworks:** React, Pytorch, .NET, Node.js, Flask, MongoDB, Ruby on Rails

**Developer Tools:** Kubernetes, Docker, Jenkins, K6, Gatling, New Relic, Snowflake, Git

## EXPERIENCE

### Software Engineer 2

July 2026 – Current

*Capital One*

*Mclean, VA*

- Building out an in house entertainment portal utilizing **AWS Cloudfront**, **API Gateway**, **Lambda**, **Typescript**, and **Go**.

### Software Engineer 1

August 2025 – July 2026

*Capital One*

*Mclean, VA*

- Developed a production-ready shadow validation service using **AWS Kinesis**, **Step Functions**, **Lambdas**, and **DynamoDB** to replay real-time production requests onto a candidate release stack, verifying release readiness and increasing release velocity by 54%.
- Developed a high-fidelity performance testing infrastructure for an internal financial core that accelerates release testing by 77%, utilizing **AWS Step Functions**, **Lambdas**, **K6**, and **Snowflake**.

### Undergraduate Student Researcher

January 2024 – May 2025

*PCVLab, The Ohio State University*

*Columbus, OH*

- Innovated MotivNet, a deep learning model for facial emotion recognition that achieved performance within 10% of state-of-the-art AffectNet FER tasks through **ViT-MAE**-based feature extraction and **Attention**-based classification.
- Conducted guided research in a highly selective undergraduate research distinction program that culminated in a published thesis, utilizing technologies such as **Pytorch**, **Weights & Biases**, and **Pandas**.
- Optimized computation by implementing distributed protocols with **SLURM**, resulting in a 43% reduction in training time.

### Software Engineering Intern

June 2024 – August 2024

*Capital One*

*Mclean, VA*

- Developed a **Kubernetes**-native controller in **Go** to mitigate cloud risk for 150+ **AWS EKS** clusters by surfacing and annotating **Docker** image vulnerabilities onto pods, accelerating the vulnerability remediation timeline by 50%.
- Designed a novel metric for an enterprise **New Relic** dashboard that identifies the vulnerability footprint of 22,500+ pods by generating vulnerability count histograms bucketed across 5 risk labels and exposing it via a **Prometheus** client.
- Optimized operator performance by at least 15% utilizing **AWS Lambda** for session management and templated operator deployment onto a CI/CD pipeline using **Jenkins** and **Helm**.

## EDUCATION

### Georgia Institute of Technology

Atlanta, Georgia

*Masters of Science in Computer Science, ML specialization*

*Spring 2026 - current*

### The Ohio State University

Columbus, OH

*Bachelor of Science in Computer Science Engineering, Minor in Business*

*Spring 2025*

Honors Research Distinction in Civil Engineering

GPA: 3.741/4.0

## PROJECTS

### K6 Open Source Contributor | [PR #5380](#) | [Go](#)

August 2025 – current

### Kubefs | [Github](#) | [Go](#), [Kubernetes](#), [Docker](#), [Helm](#)

May 2024 – September 2025

- Published a CLI tool to **Homebrew** that automates fullstack application development, testing, & deployment onto **K8** clusters by utilizing **Go**, **Cobra**, **Docker**, and **Helm**, resulting in a 23% reduction in time-to-deploy.
- Integrated support for common frameworks including **Postgres**, **Redis**, and **NextJS** to reduce operational overhead.

### Evolate | [Github](#) | [Python](#), [Pytorch](#), [Scipy](#), [Pandas](#), [Numpy](#)

May 2023 – August 2023

- Engineered a data structure that autonomously switches between different data structures and search algorithms based on user behavioral patterns, resulting in a 11% performance boost in computation speed.
- Trained a custom **PyTorch Neural Network** to determine the optimal implementations based on behavioral metrics such as insertion and deletion frequencies, search predictions, and search randomness.